

Name: _____

Date: December 23, 2019
Time allotted: 1hr.

Select campus/school: CSS | CHLSJC | CEBS | CBE-FBE | CBE-Commerce | CLGS | Art | Yared

Id no.: _____

Section: _____

General instruction: The exam contains total of 30 questions provided in three parts: true/false, multiple choice, and short answer. All answers should be written on the answer sheet included at the bottom of page 3. Switch-off mobile phone.

Part I: True or False (1 pt. each)

1. Sophia has been designed by Ethiopian Engineers and is an example of limited memory AI type.
2. When apps move from the cloud back to on-premises it is known as cloud repatriation.
3. Real-time processing is most useful when dealing with very large datasets that require quite a bit of computation.
4. Data curation involves key activities concerned with annotating data so that it becomes easily accessible, discoverable and fit for purpose.
5. Self-driving cars are example of reactive machines artificial intelligence.

Part II: Multiple Choice (1 pt. each)

1. Which of the following is not a future trend of networks?
 - a. 5G technology
 - ☒ b. Rise of centralization
 - c. Embedded computation
 - d. Network developments in edge computing
2. Which of the following is regarded as smart revolution?
 - a. Information revolution
 - b. Agriculture revolution
 - ☒ c. Industrial revolution (Second revolution)
 - d. Knowledge revolution
3. Which of the following factor is not considered as a driver of information revolution?
 - a. The growth of information-based occupation
 - b. The invention artificial intelligence
 - ☒ c. The widespread of cheap personal computers
 - d. The advent of the Internet
4. A device that provides interfacing, data communication, signal processing and other key similar functionalities is _____.
 - a. Microprocessor
 - b. Memory device
 - c. Logic device
 - ☒ d. Network device
5. Stage of industrial revolution characterized by the use of IoT, AI, and Big Data Technologies to the industry that will allow the intelligent production.
 - a. Industry 1.0
 - b. Industry 2.0
 - c. Industry 3.0
 - ☒ d. Industry 4.0

6. The bi-direction information flow between a human brain and a machine is referred to as
 - a. Neuro-modulation
 - b. Social Computing
 - c. Human Machine Interface
 - ☒ d. Brain Computer Interface
7. Agricultural mass crop production technologies were invented during ____ period
 - a. Industrial revolution
 - b. First Agricultural revolution
 - ☒ c. Second Agricultural revolution
 - d. Third Agricultural revolution
8. Poorly designed human-machine interfaces can lead to many unexpected problems. All can be classic example, except ____
 - a. Three Mile Island accident in USA
 - b. A nuclear meltdown accident
 - ☒ c. Accidents in aviation industries
 - d. Explanation of Chabots
9. All are characteristics that determine the quality of information, except ____
 - a. Timely
 - b. Accuracy
 - c. Completeness
 - d. Inconsistency
 - e. All of the above
10. The goal of most big data systems is ____
 - a. To save cost
 - b. To reduce complexity
 - c. To surface insight
 - d. To secure data
11. If you select one photo in your phone and see details like the size, type, location, and image captured date, this is ____
 - a. an integer data
 - b. a metadata
 - c. a semi-structured data
 - d. a unstructured data
12. Which of the following is wrong about data type?
 - a. A data type is an attribute of data in integers, Booleans, and floating-point number.
 - b. A data type is about the structure of data such as structured, unstructured, and semi-structured.
 - c. A data type is the structure of the data for a computer programming.
 - d. A data type is data about data in metadata
13. What are the different activities of Big-data life cycle?
 - a. Ingesting data, persisting data, computing and analyzing data, and visualizing results.
 - b. Acquisition, analysis, curation, storage, and usage.
 - c. Veracity, variability, and value.
 - d. Input, processing, and output.
14. With respect to "commuting and analyzing Big data", one of the following tool is different
 - a. Apache Storm
 - b. Apache Hadoop's MapReduce
 - c. Apache Flink
 - d. Apache Spark
15. Which of the following are the applications of Chatbots?
 - a. Google Assistance
 - b. Google Home
 - c. Siri in iphone
 - d. None of the above

16. _____ require intelligence to handle tasks such as object manipulation and navigation, along with sub-problems of localization, motion planning and mapping.
- Machine learning
 - Robotics
 - Virtual reality
 - Augmented reality
17. The most common and currently available AI system is _____
- General Intelligence
 - Narrow Intelligence
 - Super Intelligence
 - Theory of Mind
18. Which of the following does not belong to components of AI system?
- Models
 - Data
 - Hardware
 - Programming languages
19. Which of the following AI system has beaten the world's chess champion?
- AlphaGo
 - Deep Blue
 - Siri
 - Watson
20. All are AI applications in the agriculture sector, except _____
- See & Spray Robot
 - Harvest CROO Robotics
 - Crop and Soil Monitoring
 - IBM Watson analysis

Part III: Short Answer (1 pt each)

- _____ is a discipline concerned with the design, evaluation and implementation of interactive computing systems for human use and with the study of major phenomena surrounding them.
- A technology that enable machines to analyze, understand and manipulate images and video is known as _____.
- _____ is the science and engineering of making machines to think and act humanly.
- The two method of computing for Big Data Processing are _____ and _____.
- The "three Vs. of big data" are _____, _____, and _____.

Answer Sheet for Part I and Part II

Part I: True or False

1.	2.	3.	4.	5.
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Part II: Multiple Choice

1.	2.	3.	4.	5.
6.	7.	8.	9.	10.
11.	12.	13.	14.	15.
16.	17.	18.	19.	20.

Part III: Short Answer

- _____
- _____
- _____
- _____ and _____
- _____, _____, and _____